# Michael Ichikawa

AI ENGINEER · AGENTIC SYSTEMS

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## SUMMARY

I shipped a production AI agent solo, from an empty repo to live and transactable on the Microsoft Commercial Marketplace, doing the product, the engineering and the compliance myself. I built a multi-agent workflow engine from scratch for the same class of problem a level up, composing single-purpose agents into a validated graph a person signs off before anything runs. Before all of it, MS Mathematics, eight years teaching undergraduate math, and three years of semiconductor physical design at Intel.

## FLAGSHIP WORK

### Microclaw — Production AI Agent for Microsoft 365

Live on the Microsoft Marketplace

One agent across 12 Microsoft 365 services, with 85 function-calling tools and a plan/act/observe loop that chains them until a task is complete. The hard part was choosing: handing the model every tool made it worse, so a two-stage retrieval step narrows the candidates before the model ever sees them. Around that: RAG semantic memory, model routing that escalates mid-task, and multi-layer failure recovery.

*TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep*

### Multi-Agent Workflow Engine — built from scratch

mtichikawa.github.io/workflow-engine

A Python engine that turns a plain-English request into a control-flow graph of single-purpose agents, validated before it runs and gated on human approval before anything acts. The agents are a shared library reused across workflows rather than duplicated, and a learning loop improves each one from corrected examples rather than a fine-tune, taking a deliberately crippled specialist to 100 percent on a held-out eval after six exemplars. Tested on real public data in 12 unrelated domains, with bidirectional MCP interop as an edge-only adapter.

*Python · agentic orchestration · MCP (client + server) · contracts + recipes · static graph validation*

## TECHNICAL SKILLS

| | |
|---|---|
| AI-Native Dev | directing AI agents to implement, test and debug complex features · Claude Code (daily driver) · agent-directed builds verified by tests and evals · per-repo context files · AI-native development practices |
| Agents / LLM | agent orchestration · plan/act/observe loops · function calling · tool use · tool retrieval (KNN) · RAG (retrieval-augmented generation) · vector search & embeddings (pgvector · FAISS · BigQuery Vector Search) · semantic memory · multi-agent workflows · human-in-the-loop gating · MCP (client + server) · in-context learning |
| Reliability & Cost | failure-mode engineering (backoff/retry · context compaction · loop detection · abort ceilings) · model routing with mid-task escalation · input-token reduction · multi-tenant isolation (Postgres RLS) |
| Evaluation | eval design · expected-facts scoring · A/B against baselines · regression gating · pytest · Vitest · GitHub Actions CI |
| Built with | Python · TypeScript · Node.js · SQL · Azure OpenAI · Anthropic API · Microsoft Graph · PostgreSQL · BigQuery · Delta Lake (medallion) · Spark Structured Streaming · SQLAlchemy · pandas · scikit-learn · HuggingFace · sentence-transformers · Azure (Container Apps · Entra · Bicep) · AWS (S3 · Lambda · DynamoDB) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI / REST · Redis |

## EXPERIENCE

### Founding AI/ML Engineer

2026 – present

Microclaw LLC · Portland, OR

- Built the agent loop and the retrieval layer that decides which of 85 tools it may call, taking retrieval recall from 92.7 to 95.0 percent on 1,020 labelled requests against an embedding-only baseline, and the eval suite that gated every release
- Eleven weeks from first commit to a transactable Marketplace listing, solo, through Microsoft AppSource certification and the Fulfillment and Metering API work that makes it billable
- Multi-tenant isolation via Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, and delegated Entra auth

### Career Break — Full-Time Parent · Self-Directed Transition to Software

2023 – 2026

Portland, OR

- Three years as primary caregiver, retraining into software throughout
- Started in data engineering and ML, then moved to agentic development as the field did

### Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for eight years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Wrote and graded exams, and revised how I taught the topics students consistently missed
- Open-enrollment college, where one section could hold engineering majors alongside students repeating a required course

### Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm), including timing closure, power analysis, and design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run

## EDUCATION

**MS Mathematics** — The City College of New York — 2014

**BS Mechanical Engineering** — University of California, Berkeley — 2008

## PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Self-directed work from the 2023–2026 transition above, before the move to agentic systems. The one closest to current work is a **RAG pipeline on GCP** with three interchangeable retrievers behind one interface, BigQuery Vector Search, FAISS and pgvector, so they could be compared on the same eval.

The rest spans data engineering and ML: streaming and lakehouse pipelines · cloud ETL · anomaly detection · LLM tooling and bias testing · forecasting · a trading backtest system end to end.